ASTR 21200

Observational Techniques in Astrophysics
Lecture-7

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Plan for Next ~couple weeks

- Today:
  - Brief (5-10-min) intro to Lab-2, then start working in groups
- Homework-3 posted on Canvas:
  - Due Tuesday Apr-29 at 5pm
- Lab-2 Due Thursday May-7 at 5pm
  - Need to do several observations with SEO, so important to plan to observe when there are openings:
    - Note: Bad weather at SEO tonight!
  - Analysis will be more involved than previous lab, so will be important to play around how to best: a) measure galaxy sizes, and b) fit for the Hubble constant.

5	Apr-21	Intro to Lab2, More Statistics	[Lect-7]	[Lab-2, Due May-7]	
	Apr-23	(Analysis and Help/Hack Session)			
6	Apr-28	(Analysis and Help/Hack Session)			
	Apr-30	(Analysis and Help/Hack Session)			
7	May-5	Intro to Lab 3, Project Ideas	[Lect-8]	Lab-3, Due Friday May-22	
	May-7	(Analysis and Help/Hack Session)			
8	May-12	(Analysis and Help/Hack Session)			
	May-14	(Analysis and Help/Hack Session)			
9	May-19	Scientific Presentation			
	May-21	(Help / Hack, Science Presentation)			
10	May-28	Final Group Presentations: 1-3pm			

Lab-2: Measuring the Hubble Constant using “Standard Rulers”

1 Introduction

In 1929, Edwin Hubble measured that distant galaxies were systematically redshifted relative to galaxies that were closer (see Figure 1). From this data, Hubble inferred that the universe was expanding, an idea initially worked out by Georges Lemaitre using EinsteinÕs theory of gravity. In this lab, you will conduct a measurement similar to Hubble’s and will produce your own version of his famous Hubble diagram shown below.

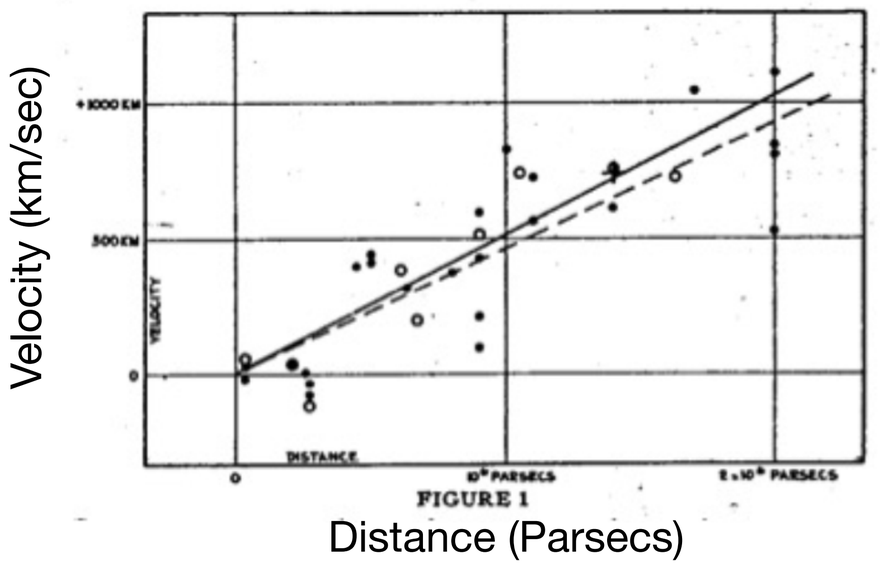


Figure 1: Velocity-Distance Relation among Extra-Galactic Nebulae (Hubble et al. 1929).

- The Universe is expanding; galaxies that are further away, also appear to be moving away from us at larger velocity.
- One way we measure this expansion is through “**Standard Rulers**” or (“**Standard Candles**”), i.e.,
  - Observe objects of **known size** (or **known luminosity**)
  - By measuring its apparent angular size (or flux) we can then infer its **distance**.
  - By also measuring its **redshift** (or recessional **velocity**), we can then measure the expansion of the Universe (or Hubble constant).
  - Typically Hubble constant measured in units of [km / s / Mpc]

Lab-2: Method

2.1 Measuring Distances

We will use geometry to measure the distance of our galaxies, by assuming that they can be used as *standard rulers*, objects with the same physical size. Galaxies that are closer will look bigger and subtend a larger angle. Whereas galaxies that are further away will look smaller and subtend a smaller angle. This relationship between the angular size of the galaxy and its distance is illustrated in Figure 3.

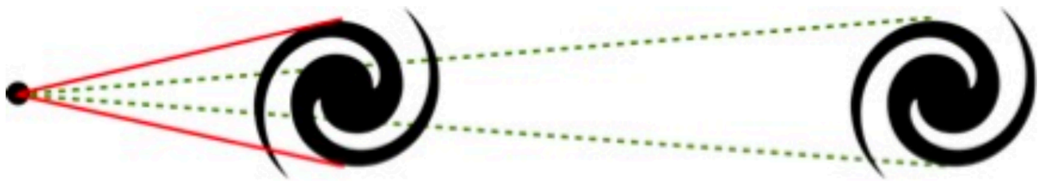


Figure 3:

Typical galaxies are about 22 kpc across. Using the geometry shown in the Figure 3, we can arrive at the following relationship:

$$\theta_{gal} = \frac{22\,\mathrm{kpc}}{d_{gal}} \tag{1}$$

where θ_{gal} is the angular size of our galaxy images in *radians* and d_{gal} is the distance to galaxies in *kpc*. So, by measuring the angular size of our galaxy images, we can use the above equation to determine the distance to the galaxy.

- Assume "galaxies" are standard rulers of fixed size.
- By assuming galaxies are a **fixed physical size** (22 kpc) and measuring their **apparent angular size**, we can infer the galaxy's distance.
- By also looking up the galaxy's "**redshift**" **(z)** on Simbad or NED, we can infer an expansion rate [velocity/ distance] from each galaxy.

Lab-2: Observations

- **Observe at least 6 galaxies between:**
  - **0.002 < redshift (z) < 0.1**
  - **6 < Magnitude (M) < 15**
- Note: Don't only pick galaxies on the lower edge of this redshift range.
  - Recessional velocity, v=c*z
  - For z=0.002, implies velocity = (3 x 10^5 km/s) (0.002) = 600 km/s
  - "***Peculiar velocities***" due to gravitational pull from local structures are typically ~200-300 km/sec, and will add variance (and maybe bias) in velocity/redshifts for nearby galaxies.

Possible Galaxies to Observe for this Lab:
Messier and NGC catalogs will be good resource to find additional galaxies to observe.

Galaxy Name	z	R.A. (J2000)	Decl. (J2000)	Notes
NGC 2775				
NGC 3034				
NGC 3147	0.0093	10:16:53.7	73:24:03	
NGC 3227				
NGC 3368				
NGC 3516				
NGC 3941				
NGC 4486				
NGC 4501	0.0076	12:31:59.2	14:25:12	
NGC 4565	0.0041	12:36:20.8	25:59:16	
NGC 4579				
NGC 4775				
NGC 5248				
NGC 5548	0.0172	14:17:59.5	25:08:12	
NGC 5746				
NGC 5907				
NGC 6015				
NGC 6181	0.0079	16:32:20.9	19:49:36	
NGC 6217				
NGC 6643				
NGC 6764				
NGC 7331				

Table 1: A list of potential galaxies that should be reasonable to observe for this lab, with space given for the redshift, Right Ascension (R.A.), and Declination (Decl.) for each object. You should choose a mix of spiral and elliptical galaxies.

Lab-2: Compare to SDSS & Legacy Survey Data

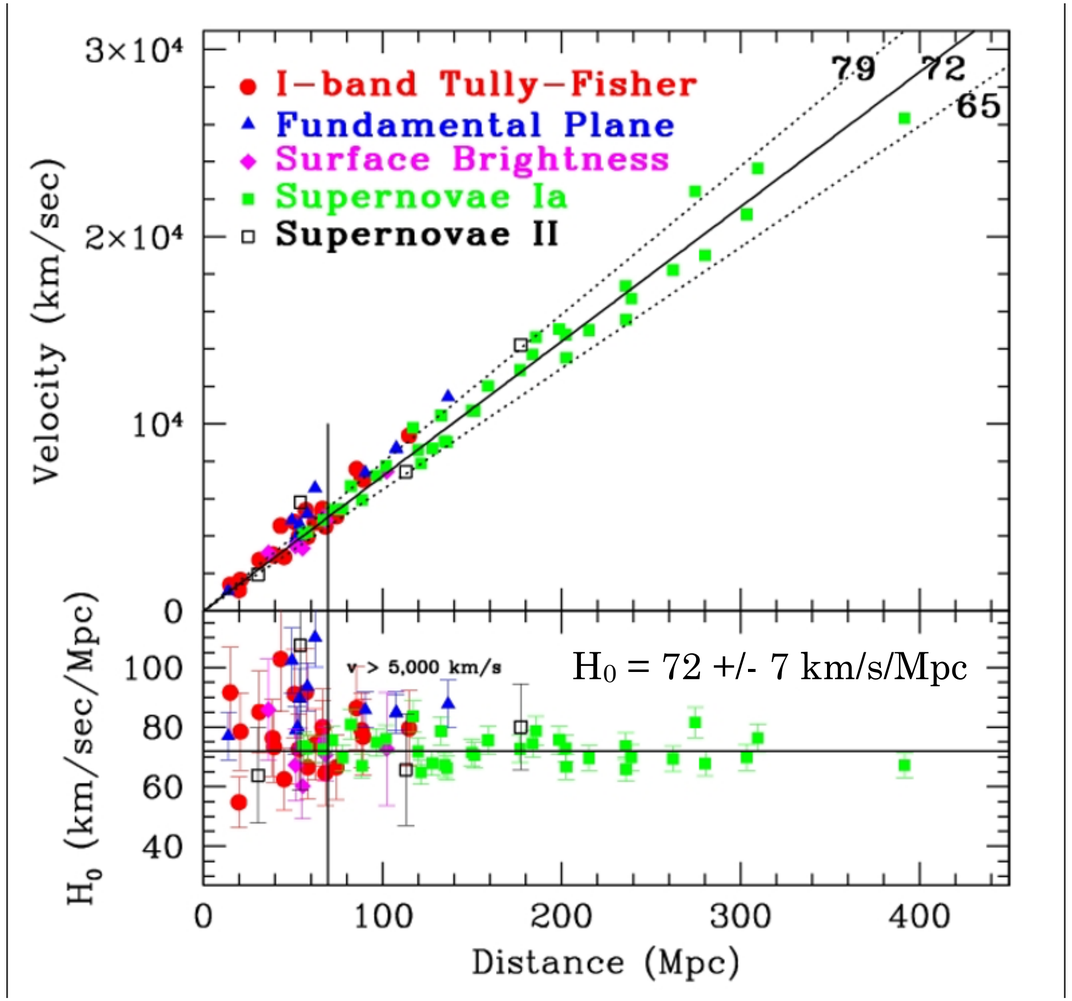
4.3 Compare to SDSS or Legacy Survey Images

There are some systematic uncertainties that could be intrinsic to our method or assumptions. However, there could also be some systematic uncertainties that could be better controlled with higher sensitivity data (e.g., if the SEO sky brightness or depth was causing a noise bias in our size measurements). To try and control for this latter type of systematic uncertainty, for at least two of your galaxies download the fits data from either the Legacy Survey (LS) or SDSS image server. As a reminder from Homework-3, these are the instructions:

1. For LegacySurvey, you can use a call like this: `https://www.legacysurvey.org/viewer/fits-cutout?ra=187.7059&dec=12.3911&width=1024&height=1024&pixscale=0.262&layer=ls-dr10&bands=gri`
2. For SDSS, to download the fits files, its probably easiest to use python, an example is on the wiki at: `https://github.com/bradfordbenson/ASTR21200_2026/wiki/SDSS-Legacy-Survey`

Figure 5: Two galaxies that you might observe for this lab. Both images are *grz* image taken from SDSS (`https://skyserver.sdss.org/`), and are 1024 by 1024 pixels across, where each pixel is 0.8-arcsec resolution. (Left) NGC 4565 is a edge-on spiral galaxy about 40 million light years away. (Right) NGC 6181 is a barred spiral galaxy about 100 million light years away.

Lab-2: Measure Hubble Law and Fit a Hubble Constant



Freedman & Madore 2010 (arXiv: 1004.1856)

- Make a measurement of Velocity-Distance Relation (aka Hubble Law)
- Fit a Hubble constant (H_0)
- Determine goodness of fit to the data, measuring a chi2 and PTE.

5 Discussion

Discuss your results, and the level of agreement with previous results. Discuss any issues or systematic uncertainties that you've identified during your analysis, and how they affected your estimate of the *Hubble* constant; be quantitative. You should include answers to the questions below in your discussion:

1. Is your best-fit *Hubble* constant a good fit to your measurement? What is the χ^2 to the fit? Can you estimate a probability to exceed (PTE) value of the fit of the model to your data?
2. How does your *Hubble* constant measurement compare to the accepted value of 70 (km / sec / Mpc)? Does it agree within the level of uncertainty (σ) of your *Hubble* constant measurement? How many σ off is the accepted value from your best-fit value?
3. Discuss differences between the statistical and systematic uncertainties that effect this measurement. For example, if we measured 100 more galaxies this way, the statistical uncertainty should decrease like $\sqrt{N_{galaxies}}$, so could we get an arbitrarily tight constraint on the Hubble constant this way? If not, what systematics limit this measurement? And how do the systematics described in Section 4.2 and 4.3, affect your measurement here?

Any questions?